



Assignment Title: Space Shooter — Midterm Project Report

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Course Teacher: MD. TANZEEM RAHAT

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No	Name	ID	Program	Signature
1	Sanjida Parven Alfe	23-50184-1	CSE	Alfe
2	Abrar Fiaz	23-50199-1	CSE	Abrar
3	Hridrita Saha Moon	23-50201-1	CSE	Moon

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	Total Marks	

Space Shooter — Midterm Project Report

1. Project Overview

- **Project Name:** Space Shooter
- **Language:** Python 3.10
- **Framework:** Pygame 2.6.1
- **Repository:** <https://github.com/Sanjida-Parven-Alfe/python-midterm-project.git>

Space Shooter is a 2D arcade-style space shooting game built with Python and Pygame. The player controls a spaceship to shoot down enemies, collect power-ups, and compete for the highest score. The game features a full CRUD-based player account system using CSV storage, an auto-fire mechanic, multi-ship formations, level progression, and a persistent high-score leaderboard.

2. Setup & Run Steps

Prerequisites:

Python 3.10 or higher is installed on your system.

Step-by-Step Setup:

1. Clone the repository: `git clone <repository-url>`
2. Navigate to folder: `cd python-midterm-project`
3. Create a virtual environment: `python -m venv .venv`
4. Activate the virtual environment:

Windows: `.venv\Scripts\activate`

Mac/Linux: `source .venv/bin/activate`

5. Install dependencies: `pip install -r requirements.txt`
6. Run the game: `python main.py`

Controls:

- **Left / Right Arrow Keys:** Move ship left / right.
- **Auto-fire:** Lasers fire automatically.
- **ESC Key:** Return to menu / Quit.

3. Feature List

Core Gameplay:

Auto-fire system: Player ship fires lasers automatically at a set interval.

Enemy spawning: Enemies appear from the top and move downward.

Collision detection: Handles laser-to-enemy and enemy-to-player collisions.

Health system: Player has 3 HP with invincibility frames.

Score tracking: 10 points for each enemy destroyed.

Level Progression:

Scaling: Difficulty increases every 100 points.

Speed: Enemies move faster as the level increases.

Ship upgrades: The ship's appearance changes every 3 levels.

Player Account System (CRUD Operations)

Create: New players register with Name and Age (Unique auto-ID).

Read: Login by entering name; view high scores.

Update: Update player name from the Settings screen.

Delete: Delete the account from the Settings screen.

Persistence: All data is saved in data.csv.

UI / Menu System

Main Menu: Start Game, High Scores, Settings, and Quit buttons.

High Scores Screen: Shows statistics, Top 5 scores, and a search bar.

4. Project File Structure

main.py: Entry point — initializes Pygame and runs loop.

game.py: Manager class — gameplay logic and HUD.

entities.py: Entity classes — Player, Enemy, Lasers.

storage.py: PlayerStorage class — CSV-based CRUD operations.

ui_manager.py: UIManager class — all menus and input.

data.csv: Player data file (Id, Name, Age, Score).

graphics/: Folder for image sprites.

audio/: Folder for sound assets.

5. Technologies & Tracks Used

Technology	Purpose
Python 3.10	Core programming language
Pygame 2.6.1	Game framework
CSV module	Data persistence (CRUD)
OOP (Classes)	Game architecture (4+ classes)
Type Hints	Code readability

6. Key OOP Concepts Used

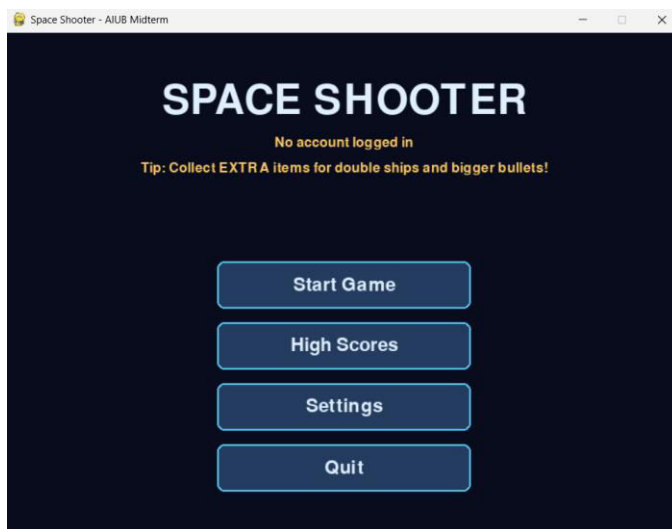
Inheritance: Player, Enemy, and Laser inherit from the Entity base class.

Encapsulation: PlayerStorage handles all CSV operations privately.

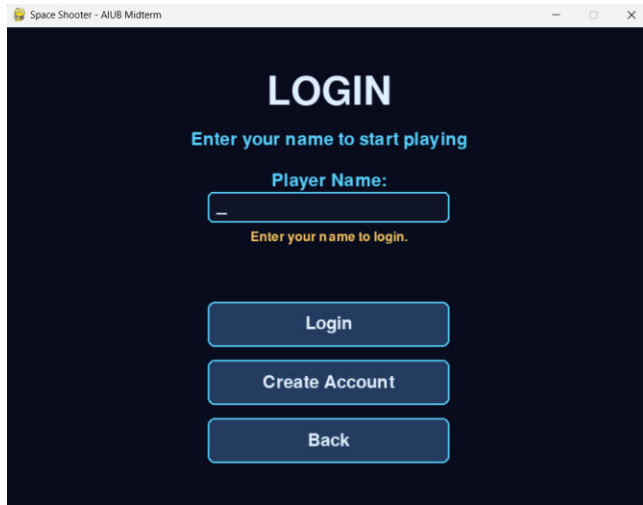
Polymorphism: Different entities use their own update() behavior.

Composition: The Game class contains and controls other classes.

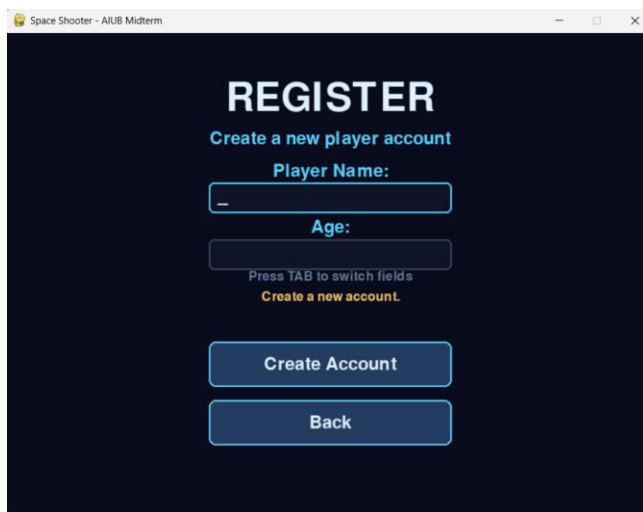
7. Screenshots & Demo Notes



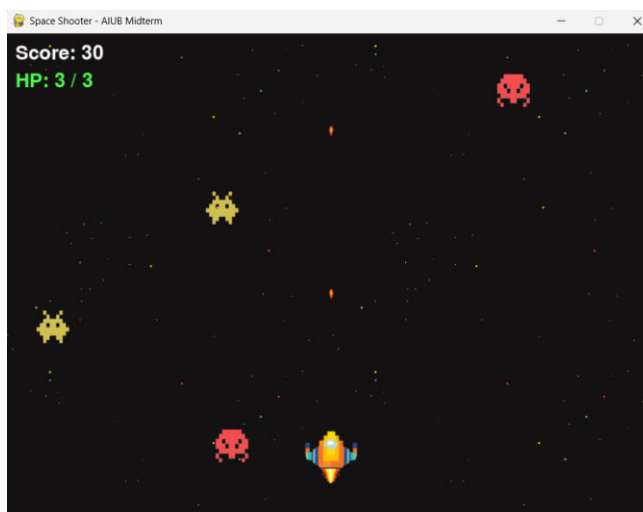
Note: "The main menu displays the game title, player login status, and navigation buttons — Start Game, High Scores, Settings, and Quit."



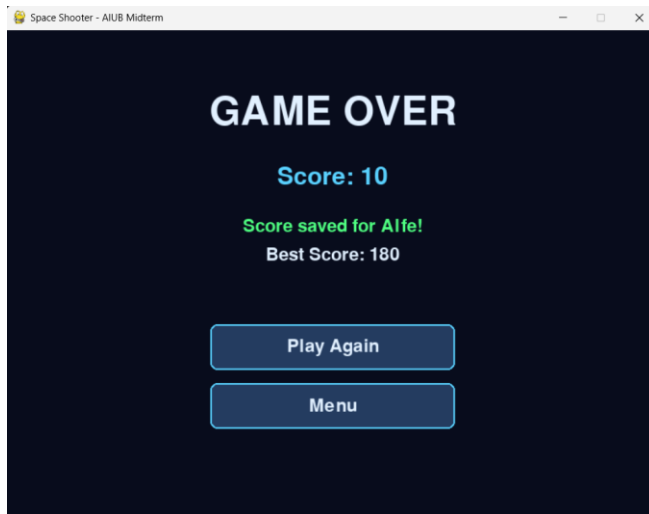
Note: "The login screen allows existing players to enter their name and start playing. This demonstrates the Read operation of CRUD — searching for a player by name in the CSV file."



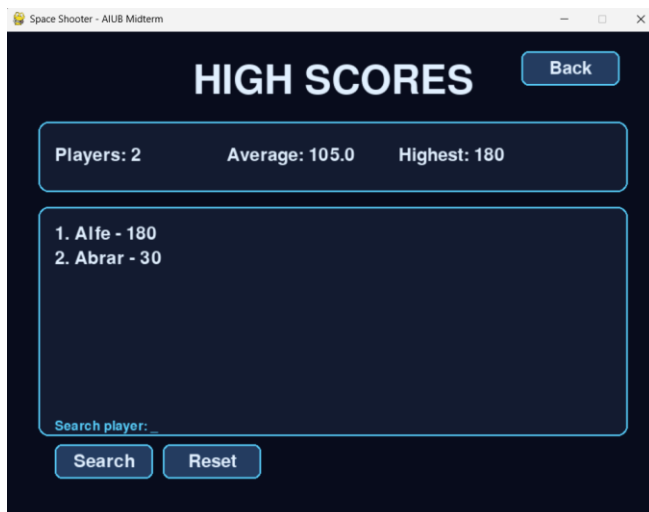
Note: "The registration screen lets new players create an account by entering their name and age. Players can press TAB to switch between input fields. This demonstrates the Create operation of CRUD — a new row is added to data.csv with an auto-generated ID."



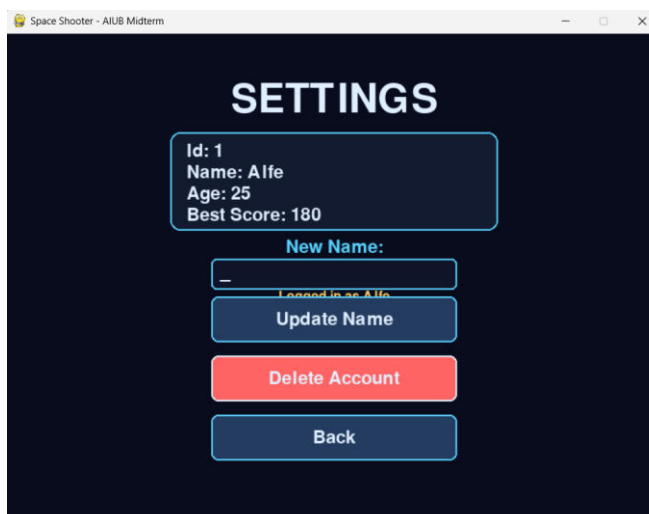
Note: "Active gameplay showing the player ship with auto-fire, incoming enemies, and the HUD displaying the current score and health (HP). The ship upgrades and enemies speed up as the level increases."



Note: "The game over screen shows the final score and confirms that the score has been automatically saved to the CSV file. The best (highest) score is always kept. No manual name input is needed — the logged-in player's score is saved automatically."



Note: "The high scores screen displays total player count, average score, highest score, and the top 5 players. A search bar allows searching for specific players by name. All data is read from data.csv."



Note: "The settings screen shows the logged-in player's information (ID, Name, Age, Best Score) and provides options to Update Name or Delete Account. This demonstrates the Update and Delete operations of CRUD."